SERCO LOADERS | GRYB

NEW OPTION · TCI EXPO 2026

JOYSTICK CONTROL FOR JIB EXTENSION

On the Serco 8500 SR, the jib now extends from a proportional thumb roller on the right-hand grip instead of a foot pedal. Same jib, same speed. Only the control moved.

SEE HOW IT WORKS →

TRY THE ROLLER

NO COST
factory option on new 8500 SR builds

SAME CYCLE
as the foot pedal

MANUAL LEVER
still runs the jib without power



SERCO 8500 SR OPERATOR STATION, RIGHT-HAND GRIP WITH THE THUMB ROLLER

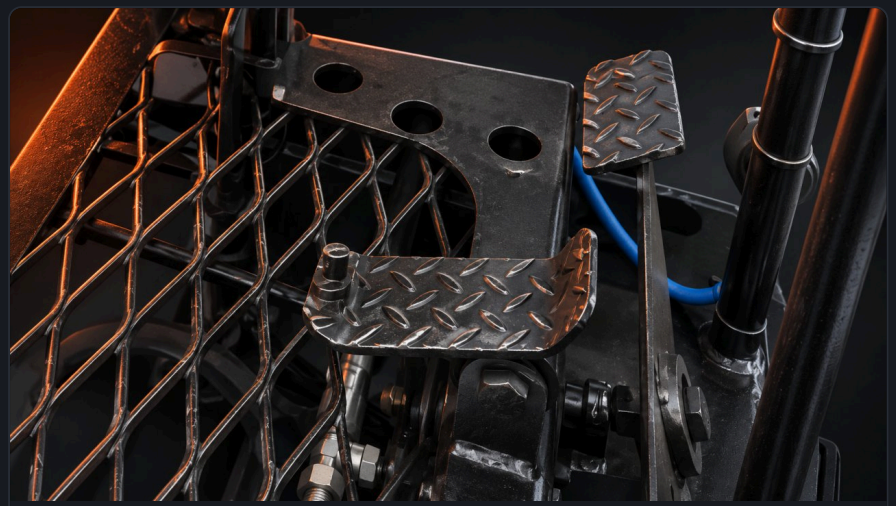
01 · WHAT CHANGED

THE JIB EXTENDS FROM A THUMB ROLLER, NOT A FOOT PEDAL.

BEFORE

FOOT PEDAL AND LINKAGES

The pedal rotates a shaft, and linkages push or pull the jib extension spool on the valve.



JIB FOOT PEDAL AND LINKAGE, SERCO 8500 · RENDERED AFTER A DEALER PHOTO

NOW

PROPORTIONAL THUMB ROLLER

A roller on the right Walvoil grip drives an electric actuator mounted on the jib extension spool. Feet stay planted, hands stay on the controls.



RIGHT-HAND WALVOIL GRIP WITH THE THUMB ROLLER · SERCO PROGRAM DOCUMENT

Same jib, same speed. Only the control moved.

02 · PROPORTIONAL CONTROL

ROLL FURTHER, FLOW MORE.

The jib moves in proportion to your thumb. Roll the wheel on the grip, drag the roller signal, or pick a position: the jib follows the factory controller curve, at the real extend and retract speeds.



8500-SR32 JIB DRIVEN BY THE ROLLER · SERCO LOADERS OFFICIAL ANIMATION

JIB POSITION 104 in of 104 in **2.8 s**

Roller forward: full extend, 104 in in 2.8 s

THUMB ROLLER



4.50 V ● EXTEND

◄ RETRACT EXTEND ►

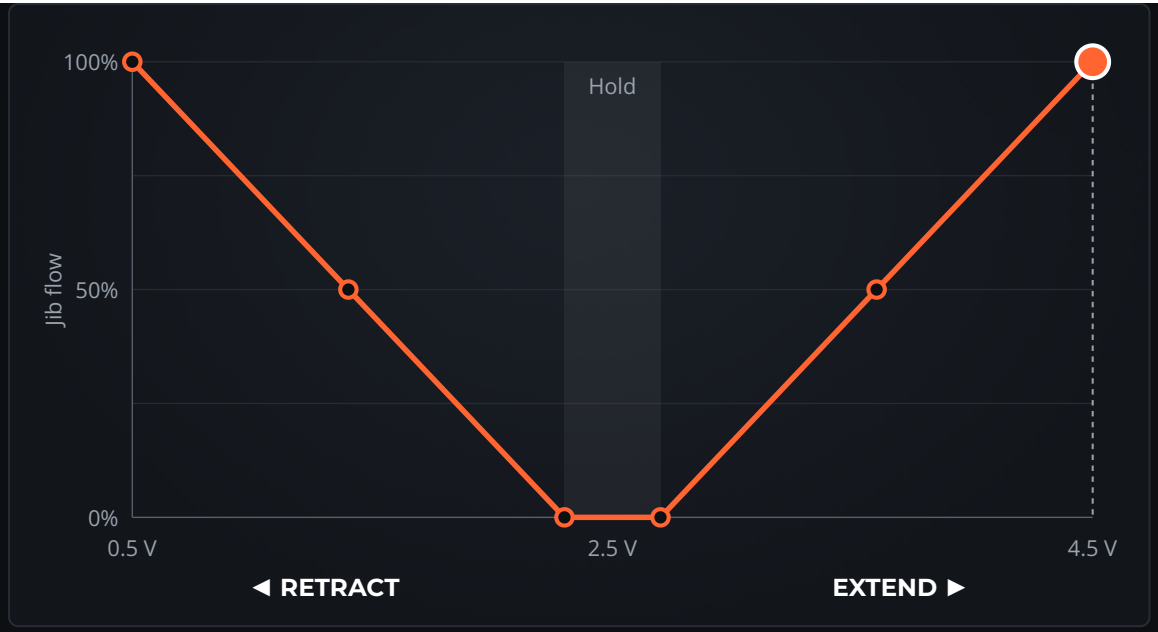
JIB FLOW

100 %

POSITIONS

0.5 V	2.5 V	4.5 V
Full retract	Neutral	Full extend

SPEC CYCLE



- **Neutral 2.5 V:** the jib holds position.
- **Center zone 2.3–2.7 V:** zero flow, so a resting thumb doesn't creep the jib.
- **Full roll:** 4.5 V gives 100% extend, 0.5 V gives 100% retract, with 25% steps in between.

Same jib hydraulics, same cycle times as the pedal.

SQUIRT STROKE	FULL EXTEND	FULL RETRACT	SYSTEM VOLTAGE
104 in	2.8 s	4.0 s	12 V

Squirt cylinder: 2.75 in bore, 2.00 in rod. The control input changes; the valve, cylinder and flow do not.

Roller signal vs. flow, factory controller program. Stroke and cycle times: Serco 8500 SR squirt function specification; jib speed shown = flow × full speed (104 in in 2.8 s extend, 4.0 s retract). Animation: Serco Loaders, 8500-SR32/SR34 animation (sercoloaders.com), positioned to follow the simulated jib.

03 · SIGNAL CHAIN

FOUR PARTS CARRY THE SIGNAL FROM THUMB TO CYLINDER.

1

THUMB ROLLER

Walvoil XMH grip, right hand. Sends 0.5–4.5 V, 2.5 V at center.

2

CONTROLLER

Axiomatic AX031700 reads the roller and sends a CAN (SAE J1939) command.

3

SPOOL ACTUATOR

Thomas EMA 46 on the D16 valve's jib section, supplied by Walvoil. Moves the spool on CAN command.

4

JIB CYLINDER

The 104-in squirt cylinder extends or retracts in proportion to the roller.

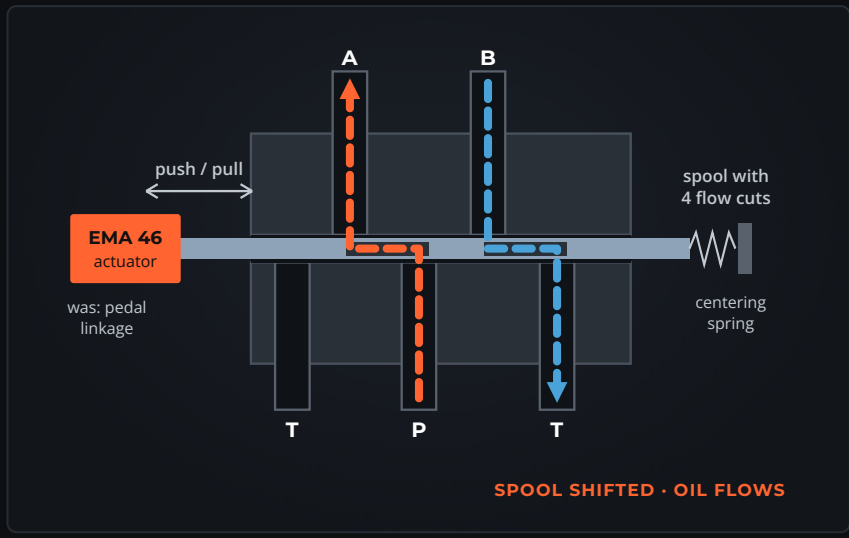
KEY-ON ▸ SELF-CHECK 2 S — ROLLER LIVE

12 V DC system, fused at 15 A after the slip ring. The controller runs a 2-second self-check at key-on before the roller goes live.

04 · ACTUATOR AND VALVE

A THOMAS EMA 46 DRIVES THE D16 JIB SPOOL DIRECTLY.

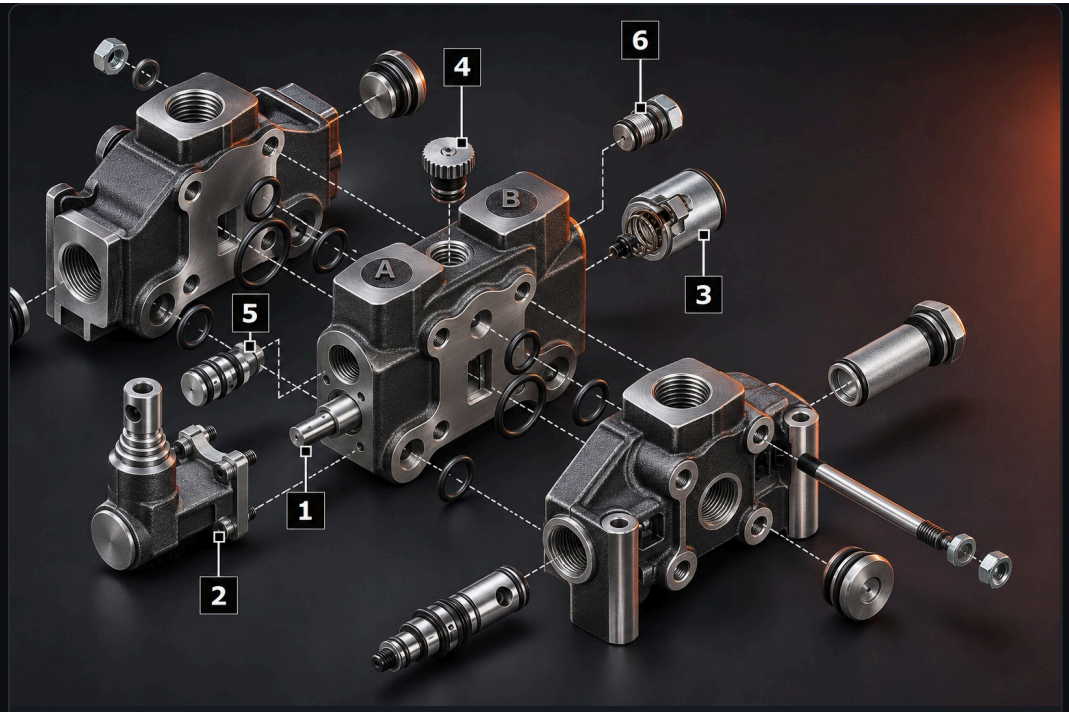




22 mm	±16 µm	IP6K6 / IPX9K
stroke	repeatability, no overshoot or oscillation	−40 to +100 °C

CAN-controlled, no pilot pressure: supplied by Walvoil, mounted on the jib section of the D16 valve.

- **Pressure P → work port:** shifting the spool opens the same gallery it always did.
- **Return port → T:** oil from the cylinder returns to tank through the same path.
- **Same section, spool and ports.** The actuator pushes and pulls the spool where the pedal linkage did: push one way to extend, pull the other way to retract.



WALVOIL D16 WORKING SECTION · RENDERED AFTER THE WALVOIL CATALOG DRAWING. (1) IS THE SPOOL THE ACTUATOR MOVES.



D16 SPOOL · RENDERED AFTER THE WALVOIL CATALOG DRAWING. ITS LANDS AND GROOVES OPEN AND CLOSE THE PORTS; THE FLAT TANG ON THE RIGHT TAKES THE ACTUATOR.

Diagram simplified, not to scale. Sources: Walvoil D16 catalog; Walvoil SD16 parts catalog; Thomas EMA 46 datasheet (06.2025).

05 · BUILT FOR THE WEATHER

EVERY PART IS SEALED FOR YEAR-ROUND OUTDOOR WORK.

COMPONENT	SUPPLIER	WEATHER PROTECTION	DURABILITY
THUMB ROLLER GRIP	Walvoil XMH, ARW roller	IP67 Roller −40 to +85 °C; grip body −20 to +60 °C	Over 3 million roller cycles
CONTROLLER	Axiomatic AX031700	IP67 Fully sealed enclosure, −40 to +85 °C	SAE J1455; MIL-STD-202G vibration and 50 g shock
SPOOL ACTUATOR	Thomas EMA 46, via Walvoil	IP6K6 / IPX9K −40 to +100 °C	Bolts directly to the D16 jib section
WIRING	Serco-engineered harness	Deutsch DT and DTM sealed connectors	Routed per Serco install standard

Three independent seal points: a leak at one cannot reach the other two.

Sources: Walvoil 2IM1000183 drawing; Axiomatic TDAX031700; Thomas EMA 46 datasheet.

06 · FAIL-SAFE

LOSE POWER, KEEP WORKING.

A manual lever still runs the jib.

NO ELECTRONICS NEEDED

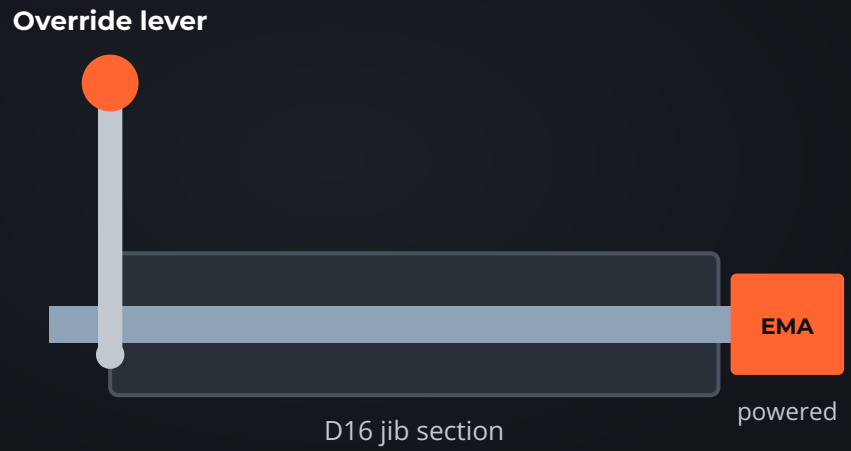
A Serco-built override lever on the jib valve section moves the spool directly. Retract the jib in under 2 minutes, no tools.

FAILS TO NEUTRAL

With power off, the spool spring-centers. Lever effort is the same as the original valve.

NEVER FIGHTS YOU

Force the lever while powered and the actuator faults and shuts off.



TRY IT

CUT POWER | PULL LEVER | RESET

Powered. The roller drives the actuator.

Schematic only. Lever position on the real valve section not shown to scale.

07 · FACTORY-SET RESPONSE

EVERY 8500 SR FEELS THE SAME.

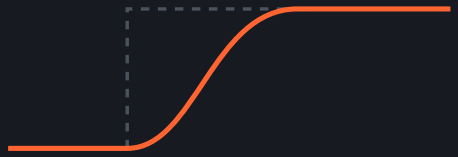
DEADBAND

A center zone sends zero flow, so small thumb movements near neutral don't move the jib.



RAMP RATE

Sets how fast the spool follows the roller, for smooth starts and stops.



MIN / MAX OUTPUT

Matched to the jib valve's travel, so full roll gives full jib speed.



All three are set by Serco at the factory. Operators don't need to adjust anything.

08 · VALIDATION

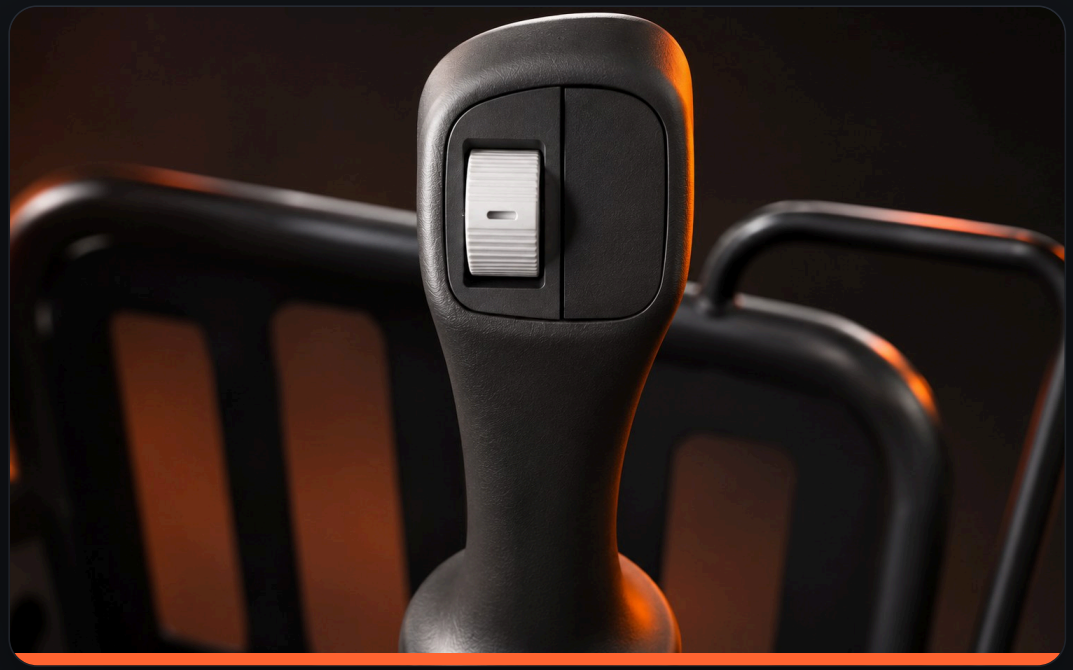
FUNCTION AND FAIL-SAFE TESTS PASSED ON A SERCO LOADER.

TEST	RESULT
Roller full forward: jib extends at full speed	✓ PASS
Roller full back: jib retracts at full speed	✓ PASS
Release the roller: jib stops	✓ PASS
Unplug the actuator: spool returns to neutral	✓ PASS
Manual lever, actuator unplugged: jib extends and retracts	✓ PASS
Plug the actuator back in: it resets	✓ PASS

Source: SpecSys Telescoping Jib Function Test SER1068 Rev A, run Sept. 2–3, 2026 at Serco.

09 · HOW TO GET IT

A NO-COST FACTORY OPTION ON NEW 8500 SR BUILDS.



PROPORTIONAL THUMB ROLLER, RIGHT-HAND GRIP

HOW TO GET IT

Available at no additional cost. Specify it when the 8500 SR is ordered; installed and tuned at the Serco plant.

WHAT STAYS THE SAME

Boom, jib in/out, grapple and rotate stay on the mechanical levers. The foot-pedal jib extension remains the standard build.

Questions: Karl Johnson, Engineering Director, Serco Loaders

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